Kernel-Based Nonlinear Anomaly Detection via Union Dictionary for Hyperspectral Images

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***Abstract*— Anomaly detection has been known to be an important issue in hyperspectral remote sensing applications. It aims to detect anomalous targets whose spectral signatures are very different from the background pixels. Although many linear detectors have obtained acceptable detection results, the linear model might not be able to describe complex hyperspectral data and could be replaced by nonlinear models. In this article, we investigate the intrinsic nonlinear characteristics of hyperspectral images (HSIs) on basis of the nonlinear mixing models and propose a novel nonlinear hyperspectral anomaly detection method based on kernel theory and union dictionary. First, the global strong anomalies in the scene and the local background pixels are utilized to construct a union dictionary. Then, a nonlinear representation-based anomaly detection model with the constructed union dictionary is designed, in which the nonlinear mixing effect of HSIs is considered. Meanwhile, the kernel theory is exploited to deal with the nonlinear interactions among the atoms in the dictionary. Finally, the anomalous level of a test pixel is determined by the representation coefficients associated with the anomaly dictionary. The proposed method is evaluated on both synthetic and real hyperspectral datasets. Experimental results demonstrate its excellent performance in comparison with linear and nonlinear state-of-the-art anomaly detectors.**

***Index Terms*— Anomaly detection, hyperspectral images (HSIs), kernel theory, nonlinear mixing models, nonlinear representation, union dictionary.**

I. Introduction

HYPERSPECTRAL images (HSIs), which can be acquired by the sensors onboard airborne or spaceborne platforms, contain rich information about the ground surface. Each pixel of the HSIs can be regarded as a spectral vector with hundreds or thousands of elements, which can be effectively used to distinguish the materials in the scene [1], [2]. This powerful discriminative capability has been exploited in many applications, such as classification and target detection [1]–[4].

Target detection for HSIs can be viewed as a binary classification performed in the pixel level, the aim of which is to determine each pixel as a target or background. The target refers to the pixels that contain specific spectral signatures, and the background is widely distributed in the HSIs [4]. According to the availability of the prior target information, target detection can be divided into supervised and unsupervised types. Specifically, the unsupervised target detection is also referred to as anomaly detection, which detects the pixels with significantly different spectra from the dominant background. Due to the fact that it is difficult to obtain the accurate spectral signatures of the targets beforehand and, furthermore, the spectral variability needs to be addressed in supervised target detection [5], anomaly detection is, therefore, more consistent with the real applications, such as search and rescue, mineral detection, and environmental surveillance [1], [6].

Many anomaly detection methods for HSIs have been proposed in the past few decades. Most of them are researched in a linear domain due to the simple physical interpretation and estimation process. The most typical method for anomaly detection is Reed–Xiaoli (RX) detector [7], which is recognized as the benchmark among the statistical approaches. It assumes that the background of HSIs follows a multivariate Gaussian distribution and estimates the mean and covariance of the background by the entire HSIs. Then, the Mahalanobis distance is computed between the test pixel and the estimated background mean to determine the probability of the test pixel being anomaly. However, the real HSIs usually do not follow a Gaussian distribution and also the process that using all data to estimate the statistics of the background is often affected by the contamination of the anomalies. As a result, several improved versions of RX detector have been proposed, including the local RX (LRX) [8], regularized RX [9], subspace RX (SSRX) [10], segmented RX [11], and iterative linear RX [12]. Apart from the statistical approaches, representation-based methods have also shown potential in anomaly detection since they can avoid the assumption about data distribution [13]–[15]. In [16], a collaborative-representation-based detector (CRD) is designed, which assumes that background pixels can be represented by the linear combination of other pixels in the surroundings, whereas anomalous pixels cannot. Moreover, a distance-weighted regularization matrix is utilized to adjust the weight of the surrounding pixels. Similarly, the background joint sparse representation detection (BJSRD) method is proposed in [17]. It applies the JSR model to select the background pixels in the dual window for representing the test pixel, and the residual error is utilized to identify the anomalous targets. CRD and BJSRD only consider the background information in the HSIs, and the pixels in

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the surroundings can be seen as the background dictionary. Moreover, it is believed that the strong anomalous pixels are significantly different from the background, and these pixels can be pre-detected and serve as prior to obtain the final detection results [18]. The detector proposed in [19] constructs the union dictionary with the background and potential anomalous information to represent each test pixel, which can enhance the representation ability and improve the detection performance. However, these methods only consider the linear combination. In reality, there are many situations where the linear methods are not adequate to model the real HSIs due to the intrinsic nonlinear characteristics [20]. In these cases, nonlinear models should be taken into consideration.

In order to design the nonlinear anomaly detectors, manifold learning and kernel methods are two kinds of effective approaches. Manifold learning deals with the nonlinear characteristics of the HSIs by considering that the data are embedded in an underlying low-dimensional manifold space. Manifold ranking-based detection framework (MRDF) [18] utilizes the manifold ranking to generate the detection map. Besides, the robust background regression-based score estimation (RBRSE) [21] exploits the manifold smoothness of the HSIs for anomaly detection. Different from manifold learning, kernel methods map the data into a high-dimensional feature space and can extend the linear detector to its nonlinear version [22]–[24]. Kernel RX (KRX) [25] is a nonlinear version of RX by using the kernel theory to implement the nonlinear mapping. The nonlinear features extracted in the high-dimensional space can enhance the separability and discrimination of the data. In [16], the proposed CRD is also extended to a kernel version, referred to as kernel CRD (KCRD). Similar to the KRX, the KCRD maps each pixel into a high-dimensional feature space for the collaborative representation, which is implicitly achieved by the kernel function. Besides, the nonlinear anomaly detector based on support vector data description (SVDD) [26] is a single-class classifier that models the background pixels in a minimum enclosing hypersphere, and those falling outside of this hypersphere are determined as the anomalies. The aforementioned nonlinear methods simply apply the nonlinear mapping on each pixel to extract the nonlinear features of the spectral bands. Although this strategy can increase the number of features and enhance the separability of the data and the detectability of anomalies, it is not related to the intrinsic nonlinear characteristics in HSIs. Moreover, the inappropriate nonlinear mapping may lead to the high false alarm in the detection results, and also, there is no general method for the selection of the kernel function [27].

Specifically, the linear inseparability between the anomaly and the background can be attributed to the nonlinear mixing effect of the hyperspectral data [28], [29]. Due to the low spatial resolution of the HSIs, each pixel may consist of several types of material spectra (endmembers), and the nonlinear mixture is common in the natural scenes [30]–[33]. For example, multiple scattering effects can be observed in urban areas [34]. Another typical situation is that light usually interacts with several materials simultaneously in an intimate association [35]. According to the hyperspectral nonlinear mixing models, such as the bilinear mixing models (BMMs) (including the polynomial post-nonlinear model (PPNM) [36]) and the multilinear mixing (MLM) [37] model, the nonlinear interactions of HSIs exist in the endmembers rather than the spectral bands. Therefore, it is much more meaningful to pay close attention to the nonlinear interactions of the pixels instead of the individual spectral distortions for nonlinear anomaly detection.

In this article, we propose a novel kernel-based nonlinear anomaly detection method via union dictionary (KNUD) for HSIs. The proposed method is a representation-based detector, and each test pixel is assumed as the combination of the atoms in the union dictionary. Specifically, on basis of the nonlinear mixing models, the combination consisting of linear and nonlinear parts can model the nonlinear interactions of the atoms in the dictionary. Unlike the CRD or BJSRD that constructs the dictionary with background-only pixels, the proposed detector constructs the union dictionary (including background dictionary and potential anomaly dictionary) to model the local spatial correlation and global anomaly information simultaneously. We exploit the dual window to construct the background dictionary. For the potential anomaly dictionary, some simple detectors, such as global RX (GRX) [7], can be utilized to obtain the initial detection map and the strong anomalous pixels can be easily picked out and constitute the anomaly dictionary. Then, a specific optimization problem is formulated with respect to the representation coefficients associated with the constructed union dictionary for each test pixel. Once the representation problem is solved, the representation coefficient vector of the anomaly dictionary is used to identify the anomalies.

The main contributions of this article can be concisely summarized as follows.

1) A novel nonlinear representation-based anomaly detection method KNUD is proposed for HSIs. As far as we know, it is the first time to take nonlinear interactions among the dictionary atoms into account for hyperspectral anomaly detection. With the analysis of intrinsic nonlinear characteristics of HSIs based on the nonlinear mixing theory, the proposed KNUD is much more physically interpretable than those that simply extract the nonlinear features by the individual spectral distortions.
2) Both the global anomalous information and the local spatial correlations are considered in the construction process of the union dictionary. To represent each test pixel via the constructed union dictionary, an optimization problem is specifically formulated over the representation coefficients. According to the representation coefficients of the potential anomaly dictionary, the anomalous pixels of the HSIs can be identified accurately.

The remainder of this article is organized as follows. Section II reviews the related works. The details of the proposed anomaly detection method are presented in Section III. In Section IV, the experimental results on both synthetic and real HSIs are reported and discussed. Finally, we conclude this article in Section V.

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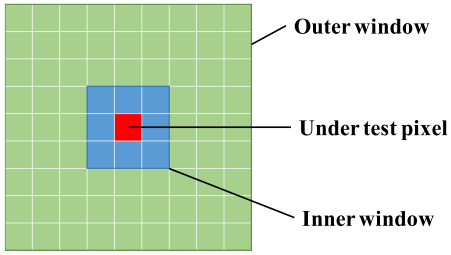


Fig. 1. Example of the dual window.

II. Related Works

This section reviews the typical representation-based detector CRD and two representative hyperspectral nonlinear mixing models (PPNM and MLM models). Their intrinsic principles and strategies play the foundational role in the proposed KNUD.

A. CRD

CRD is a typical anomaly detector based on representation theory without the assumption about the distribution of the background. It adopts the sliding dual window (see an example in Fig. 1) to estimate the background information. For each test pixel $\mathbf{y} \in \mathbb{R}^{L\times 1}$ (L is the number of the spectral bands), a concentric dual window can be formed. The pixels between the inner window and the outer window are selected to constitute a 2-D matrix $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^{K}$, where K is the number of the chosen pixels in the surrounding background. The objective of the CRD is to find the weight vector $\boldsymbol{\alpha} \in \mathbb{R}^{K}$ for the background matrix $\mathbf{X}$ to represent the test pixel under the constraint of ℓ_2-norm. The objective function is presented as

$$\arg\min_{\boldsymbol{\alpha}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\alpha}\|_2^2 + \lambda\|\boldsymbol{\alpha}\|_2^2 \tag{1}$$

where λ is a parameter to control the penalty of the constraint.

Considering the contamination in the dual window, a distance-weighted Tikhonov regularization is introduced to adjust the weight vector with the following form:

$$\boldsymbol{\Gamma}_{\mathbf{y}} = \begin{bmatrix} \|\mathbf{y} - \mathbf{x}_1\|_2 & & 0 \\ & \ddots & \\ 0 & & \|\mathbf{y} - \mathbf{x}_K\|_2 \end{bmatrix}. \tag{2}$$

In order to impose the sum-to-one constraint to representation coefficients, the optimization problem in (1) is modified as

$$\arg\min_{\hat{\boldsymbol{\alpha}}} \left\|\tilde{\mathbf{y}} - \tilde{\boldsymbol{X}}\hat{\boldsymbol{\alpha}}\right\|_2^2 + \lambda\left\|\boldsymbol{\Gamma}_{\mathbf{y}}\hat{\boldsymbol{\alpha}}\right\|_2^2 \tag{3}$$

where $\tilde{\mathbf{y}} = \begin{bmatrix}\mathbf{y}; 1\end{bmatrix}$ and $\tilde{\mathbf{X}} = [\mathbf{X}; \mathbf{1}]$.

Once the representation process is accomplished, the residual error $\|\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\alpha}}\|_2$ is used to determine the anomalies.

Furthermore, the CRD can be extended to its nonlinear version, i.e., KCRD, in which the raw pixels are mapped into a high-dimensional feature space in order to improve the data separability. Then, the optimization problem in (3) becomes

$$\arg\min_{\boldsymbol{\alpha}^*} \left\|\Phi(\mathbf{y}) - \boldsymbol{\Phi}\boldsymbol{\alpha}^*\right\|_2^2 + \lambda\left\|\boldsymbol{\Gamma}_{\Phi(\mathbf{y})}\boldsymbol{\alpha}^*\right\|_2^2 \tag{4}$$

where Φ is the nonlinear mapping: $\mathbf{y} \to \Phi(\mathbf{y})$, and $\boldsymbol{\Phi} = [\Phi(\mathbf{x}_1), \Phi(\mathbf{x}_2), \ldots, \Phi(\mathbf{x}_K)]$. The Tikhonov matrix $\boldsymbol{\Gamma}_{\Phi(\mathbf{y})}$ has the following form:

$$\boldsymbol{\Gamma}_{\Phi(\mathbf{y})} = \begin{bmatrix} \|\Phi(\mathbf{y}) - \Phi(\mathbf{x}_1)\|_2 & & 0 \\ & \ddots & \\ 0 & & \|\Phi(\mathbf{y}) - \Phi(\mathbf{x}_K)\|_2 \end{bmatrix}. \tag{5}$$

The representation residual of the KCRD can be computed by the kernel trick [24] as

$$\begin{aligned} r &= \left\|\Phi(\mathbf{y}) - \boldsymbol{\Phi}\boldsymbol{\alpha}^*\right\|_2 \\ &= \sqrt{(\Phi(\mathbf{y}) - \boldsymbol{\Phi}\boldsymbol{\alpha}^*)^T(\Phi(\mathbf{y}) - \boldsymbol{\Phi}\boldsymbol{\alpha}^*)} \\ &= \sqrt{k(\mathbf{y},\mathbf{y}) + \boldsymbol{\alpha}^{*T}\mathbf{K}\alpha^* - 2\boldsymbol{\alpha}^{*T}\mathbf{k}(\cdot,\mathbf{y})}. \end{aligned} \tag{6}$$

B. Hyperspectral Nonlinear Mixing Models

The limitation of the low spatial resolution in HSIs generally results in the fact that the region of one pixel covers several endmembers, and here, each endmember represents a type of material on the ground surface [30]. In fact, the nonlinear interactions of endmembers are common due to the multiple light scattering effects and several materials in an area with an intimate mixture. Several nonlinear mixture models have been developed to interpret the nonlinear mixing in HSIs [38], such as the Fan model (FM) [39], generalized bilinear model (GBM) [40], PPNM, and MLM. Considering the complexity and effectiveness to model the nonlinear mixing effect, we focus on PPNM and MLM models to explore the specific nonlinear interactions among the endmembers and utilize their explicit physical interpretations for anomaly detection.

PPNM only considers the second-order scattering effect to model the nonlinear parts of the real HSIs [36]. Assume that $\mathbf{M} = \begin{bmatrix}\mathbf{m}_1, \mathbf{m}_2, \ldots, \mathbf{m}_Q\end{bmatrix} \in \mathbb{R}^{L\times Q}$ is the endmember matrix and $\mathbf{x} \in \mathbb{R}^{L\times 1}$ is a pixel in HSIs, and the form of PPNM is as follows:

$$\mathbf{x} = \sum_{i=1}^{Q}\mathbf{m}_i\alpha_i + \xi\sum_{i=1}^{Q}\sum_{j=1}^{Q}(\mathbf{m}_i \odot \mathbf{m}_j)\alpha_i\alpha_j + \mathbf{n} \tag{7}$$

where $\odot$ represents the Hadamard product, $\mathbf{m}_i \odot \mathbf{m}_j = \begin{bmatrix}m_{i,1}m_{j,1}, m_{i,2}m_{j,2}, \ldots, m_{i,L}m_{j,L}\end{bmatrix}^T$ models the second-order scattering between the two endmembers, $\boldsymbol{\alpha} = \begin{bmatrix}\alpha_1, \alpha_2, \ldots, \alpha_Q\end{bmatrix}^T$ is the abundance vector associated with the pixel $\mathbf{x}$, and ξ denotes the nonlinear parameter.

It is reasonable for PPNM because the nonlinear contributions in HSIs are commonly dominated by the second-order scattering. To take the higher-order scattering into consideration, PPNM is extended to the MLM model [37], and a probability parameter P about the multiple interactions is introduced, which gives rise to

$$\begin{aligned} \mathbf{x} = {} & (1-P)\sum_{i=1}^{Q}\mathbf{m}_i\alpha_i + (1-P)P\sum_{i=1}^{Q}\sum_{j=1}^{Q}(\mathbf{m}_i \odot \mathbf{m}_j)\alpha_i\alpha_j \\ & + (1-P)P^2\sum_{i=1}^{Q}\sum_{j=1}^{Q}\sum_{k=1}^{Q}(\mathbf{m}_i \odot \mathbf{m}_j \odot \mathbf{m}_k)\alpha_i\alpha_j\alpha_k \\ & + \cdots + \mathbf{n}. \end{aligned} \tag{8}$$

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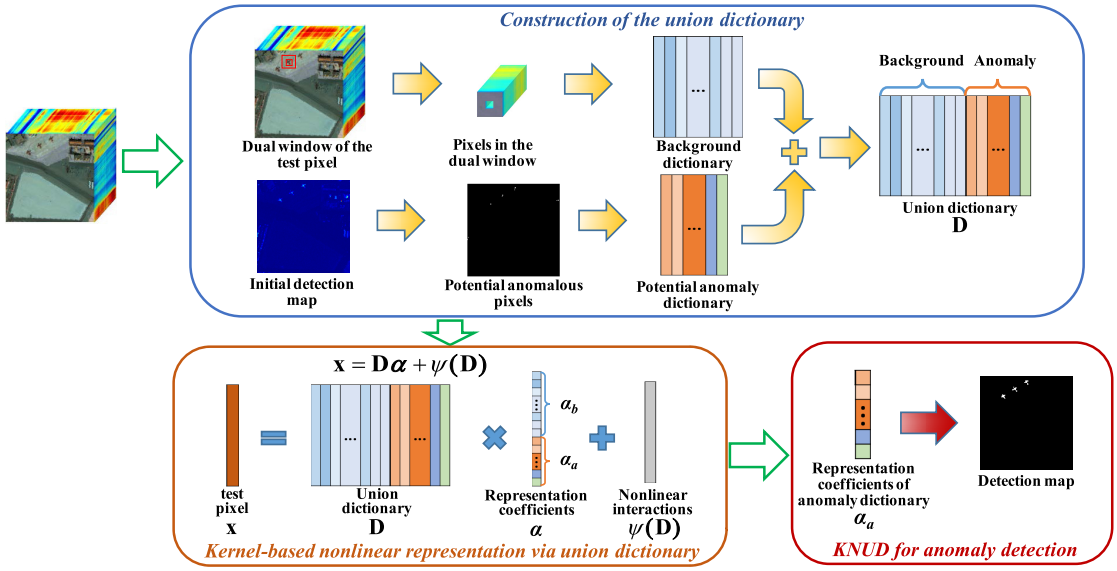


Fig. 2. Flowchart of the proposed method for hyperspectral anomaly detection.

Both the PPNM and the MLM models indicate that the nonlinear interactions among the endmembers need to be considered. Thus, for anomaly detection, it is of significant importance to consider the nonlinear interactions among the pixels rather than spectral bands.

III. Proposed Method

In this section, we describe the proposed KNUD in detail. The union dictionary is constructed by considering the local spatial correlation and the global anomaly information. Then, each test pixel is represented as the nonlinear combination of the obtained union dictionary on basis of the nonlinear mixing model. By means of the kernel theory, the nonlinear interactions among the dictionary atoms are implicitly expressed. With a specifically designed optimization procedure, the representation vector of the union dictionary can be obtained, and then, the representation coefficients of the potential anomaly dictionary are employed to determine the anomalous level. The flowchart of the proposed KNUD is shown in Fig. 2.

A. Construction of the Union Dictionary

Unlike supervised target detection, there is no concrete spectral information about the anomaly or background for anomaly detection. The union dictionary is generally constructed by the potential prior like that the spectra of the anomalous pixels are significantly different from those of the background, and the background pixels have spatial homogeneity [41]. Some anomaly detectors construct the background dictionary by clustering methods, such as k-means [42] or its extended version [19], mean shift [43], self-organizing map (SOM) [44], and density-based spatial clustering of applications with noise (DBSCAN) [45]. All pixels are grouped into several clusters and some representative pixels in each cluster are selected to construct the background dictionary. However, the detection performance is sensitive to the number of clusters. Besides, different clusters usually have very different numbers of pixels on the account of the distribution of the surface materials. It is difficult to choose the appropriate parameters for the construction of the background dictionary.

To avoid the drawback of the clustering-based methods in constructing the background dictionary, we adopt the dual window to extract the local background information, which has been successfully utilized in several anomaly detectors [8], [16], [46]. On the one hand, there are only two parameters (inner window size W_{in} and outer window size W_{out}) to locate the dual window, and satisfactory detection results can be obtained in a relatively wide range of the parameters [46]. On the other hand, the dual window is located in relation to each test pixel, which takes the local spatial correlation into account. Generally, the inner window size is larger than the probable anomaly, and the area between the inner window and the outer window is assumed as the local background region. The pixels within the local background region can be utilized to construct the background dictionary. Considering the situation that several anomalies and sparse background pixels may appear in the local background region, it will result in the contamination of the background dictionary. Thus, a distance-based strategy is employed to eliminate the interference of the anomalies. Similar to the RX detector, we compute the Mahalanobis distance between each pixel and the mean of all pixels in the local background region. A large distance generally indicates that this pixel is out of the local spatial homogeneity. According to the distance value, we rank the pixels in the local background region in an ascending order and select the first $\eta\%$ of the pixels as the atoms of the background dictionary.

For the construction of the potential anomaly dictionary, the strong anomalous pixels with a significant difference from the background can be detected by the global anomaly detectors, and these pixels can be considered as the prior to obtain final more accurate detection results [18], [19], [47]. Specifically, we utilize the GRX detector to obtain the initial detection map, anomalous probability of each pixel is ranked

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in a descending order, and the first R pixels are chosen as the potential anomaly dictionary atoms.

B. Kernel-Based Nonlinear Representation via Union Dictionary

Let $\mathbf{x} = [x_1, x_2, \ldots, x_L]^T \in \mathbb{R}^{L\times 1}$ denotes each pixel of the given HSIs, and the representation-based theory assumes that this pixel can be represented as the combination of the atoms in the union dictionary. We use $\mathbf{D}_b = \left[\mathbf{d}_{b_1}, \mathbf{d}_{b_2}, \ldots, \mathbf{d}_{b_M}\right] \in \mathbb{R}^{L\times M}$ and $\mathbf{D}_a = \left[\mathbf{d}_{a_1}, \mathbf{d}_{a_2}, \ldots, \mathbf{d}_{a_R}\right] \in \mathbb{R}^{L\times R}$ to stand for the background dictionary and the potential anomaly dictionary, respectively. In this way, the union dictionary has the form of $\mathbf{D} = [\mathbf{D}_b, \mathbf{D}_a] = \left[\mathbf{d}_{b_1}, \mathbf{d}_{b_2}, \ldots, \mathbf{d}_{b_M}, \mathbf{d}_{a_1}, \mathbf{d}_{a_2}, \ldots, \mathbf{d}_{a_R}\right] = [\mathbf{d}_1, \mathbf{d}_2, \ldots, \mathbf{d}_M, \mathbf{d}_{M+1}, \ldots, \mathbf{d}_{M+R}] \in \mathbb{R}^{L\times N}$, where M is the number of the atoms in the background dictionary, R is the number of atoms in the potential anomaly dictionary, N is the number of all atoms in the union dictionary, and $N = M + R$. For the convenience of the following expression, we use $\mathbf{d}_{\lambda_l}^T \in \mathbb{R}^{1\times N}$ denotes the l-th row of $\mathbf{D}$, that is, the spectral signatures of the atoms in the union dictionary at the l-th band.

Then, the linear representation model can be written as

$$\mathbf{x} = \mathbf{D}\boldsymbol{\alpha} + \mathbf{n} \tag{9}$$

where $\boldsymbol{\alpha} = \left[\boldsymbol{\alpha}_b^T,\ \boldsymbol{\alpha}_a^T\right]^T$ is the representation coefficient vector of the union dictionary.

Considering the nonlinear representation and the nonlinear interactions among the atoms in the dictionary, (9) can be modified by an extra nonlinear part as

$$\mathbf{x} = \mathbf{D}\boldsymbol{\alpha} + \psi(\mathbf{D}) + \mathbf{n} \tag{10}$$

where ψ is an unknown nonlinear function describing the nonlinear interactions for each spectral band in the union dictionary and $\psi(\mathbf{D}) \in \mathbb{R}^{L\times 1}$ is the nonlinear representation part with the same size as the pixel. Specifically, on account of the nonlinear mixing models, the nonlinear interactions can be modeled as the Hadamard product of the pixels, which has the influence on each spectral band of the dictionary atoms. As a result, we model the nonlinear part based on the spectral band of the union dictionary $\mathbf{d}_{\lambda_l}$.

According to the Riesz representation theorem [48], the result of the nonlinear function ψ for each fixed $\mathbf{d}_{\lambda_l}$ can be computed by a definite reproducing kernel function as

$$\psi\left(\mathbf{d}_{\lambda_l}\right) = \left\langle \psi,\ k_{\text{nlin}}\left(\cdot,\ \mathbf{d}_{\lambda_l}\right)\right\rangle_{\mathcal{H}} \tag{11}$$

where $k_{\text{nlin}}(\cdot,\ \cdot)$ is the selected kernel function. The nonlinear function ψ can also be represented by the combination of other spectral band with $k_{\text{nlin}}(\cdot,\ \mathbf{d}_{\lambda_i})$. A detailed proof of this can be found in [48]. Thus, the nonlinear part can be characterized by a proper kernel function, which is important to describe the nonlinear interactions. For the homogeneous polynomial kernel of degree q, that is, $k_{\text{nlin}}(\mathbf{d}_{\lambda_i},\ \mathbf{d}_{\lambda_j}) = (\mathbf{d}_{\lambda_i}^T\mathbf{d}_{\lambda_j})^q$, the inner product is computed in the nonlinear feature space

$$k_{\text{nlin}}\left(\mathbf{d}_{\lambda_i}, \mathbf{d}_{\lambda_j}\right) = \boldsymbol{\Phi}_{\text{nlin}}\left(\mathbf{d}_{\lambda_i}\right)^T \cdot \boldsymbol{\Phi}_{\text{nlin}}\left(\mathbf{d}_{\lambda_j}\right) \tag{12}$$

where $\boldsymbol{\Phi}_{\text{nlin}}$ is the specific feature map, which implements the interactions among the elements in the vector to construct the nonlinear features for $\mathbf{d}_{\lambda_i}$ and $\mathbf{d}_{\lambda_j}$. Actually, $(\mathbf{d}_{\lambda_i}^T\mathbf{d}_{\lambda_j})^q = (d_{\lambda_{i,1}}d_{\lambda_{j,1}} + d_{\lambda_{i,2}}d_{\lambda_{j,2}} + \cdots + d_{\lambda_{i,N}}d_{\lambda_{j,N}})^q$ can be expanded as a sum of the weighted q-th degree monomials, and each monomial has the form as

$$\left(d_{\lambda_{i,1}}d_{\lambda_{j,1}}\right)^{q_1}\left(d_{\lambda_{i,2}}d_{\lambda_{j,2}}\right)^{q_2}\ldots\left(d_{\lambda_{i,N}}d_{\lambda_{j,N}}\right)^{q_N} \tag{13}$$

where $\sum_{n=1}^{N} q_n = q$ and $q_n (n = 1, \ldots, N)$ is a nonnegative integer. This can be viewed as all possible q-order interactions. In other words, (13) can be further written as the product of the nonlinear features $d_{\lambda_{i,1}}^{q_1}d_{\lambda_{i,2}}^{q_2}\ldots d_{\lambda_{i,N}}^{q_N}$ and $d_{\lambda_{j,1}}^{q_1}d_{\lambda_{j,2}}^{q_2}\ldots d_{\lambda_{j,N}}^{q_N}$. Due to the form of the nonlinear features that are the same in each spectral band, the nonlinear interaction of the atoms in the union dictionary has the Hadamard-like form [24] as

$$\mathbf{d}_1^{q_1} \odot \mathbf{d}_2^{q_2} \odot \cdots \odot \mathbf{d}_N^{q_N}. \tag{14}$$

In this way, nonlinear interactions occur between the background and the potential anomalies.

Compared with the other nonlinear detectors that map the pixels into a high-dimensional space, the nonlinear interactions among the pixels described above are more physically interpretable to model the intrinsic nonlinear characteristics in HSIs on basis of the nonlinear mixing theory. As a result, the proposed method can better detect the anomalies in the HSIs, especially the anomalies in sub-pixel level.

For the second-order polynomial kernel

$$k_{\text{nlin}}\left(\mathbf{d}_{\lambda_i}, \mathbf{d}_{\lambda_j}\right) = \left(\mathbf{d}_{\lambda_i}^T\mathbf{d}_{\lambda_j} + 1\right)^2 \tag{15}$$

it has the ability to represent the second-order interactions of the dictionary atoms such as $\mathbf{d}_i \odot \mathbf{d}_j$ according to (14). Therefore, we can utilize the second-order polynomial kernel to detect the anomalies in HSIs that are mainly influenced by the second-order scatterings. Note that here the PPNM considers only the second-order nonlinear interactions between endmembers.

As for the Gaussian kernel with bandwidth σ

$$k_{\text{nlin}}\left(\mathbf{d}_{\lambda_i}, \mathbf{d}_{\lambda_j}\right) = \exp\left(-\frac{\left\|\mathbf{d}_{\lambda_i} - \mathbf{d}_{\lambda_j}\right\|^2}{2\sigma^2}\right) \tag{16}$$

it can be expressed as an infinite series of higher-order polynomial kernels [49]

$$k_{\text{nlin}}\left(\mathbf{d}_{\lambda_i}, \mathbf{d}_{\lambda_j}\right) = \sum_{k=0}^{\infty}\frac{\left(\mathbf{d}_{\lambda_i}^T\mathbf{d}_{\lambda_j}\right)^k}{\sigma^{2k}k!} \times \exp\left(-\frac{\left\|\mathbf{d}_{\lambda_i}\right\|^2}{2\sigma^2}\right)\exp\left(-\frac{\left\|\mathbf{d}_{\lambda_j}\right\|^2}{2\sigma^2}\right). \tag{17}$$

Moreover, it is noteworthy that MLM considers the higher-order nonlinear interactions between endmembers. Due to that the higher-order nonlinear interactions among the pixels are incorporated into the Gaussian kernel function, the proposed KNUD with Gaussian kernel may perform better in the situations that are affected by the high-order multiple-layer scattering.

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C. KNUD for Anomaly Detection

Under the assumption of the representation-based anomaly detector, the objective function of the proposed KNUD is formulated by minimizing the representation error with a regularization term, i.e.,

$$\begin{aligned}&\arg\min_{\boldsymbol{\alpha},\psi}\frac{1}{2}\sum_{l=1}^{L}e_l^2+\frac{\lambda}{2}\left(\|\boldsymbol{\alpha}\|^2+\|\psi\|_{\mathcal{H}}^2\right)\\&\text{s.t. } e_l=x_l-\mathbf{d}_{\lambda_l}^T\boldsymbol{\alpha}-\psi\left(\mathbf{d}_{\lambda_l}\right)\\&\quad\boldsymbol{\alpha}\geq 0\quad\text{and }\mathbf{1}^T\boldsymbol{\alpha}=1\end{aligned}\tag{18}$$

where $\mathcal{H}$ is a given functional space and λ is a positive parameter to control the tradeoff between the representation error and the regularization term.

This convex optimization prroblem can be solved by its dual problem according to the strong duality property [50]. The Lagrange function associated with (18) is formulated as

$$\begin{aligned}L=&\frac{1}{2}\sum_{l=1}^{L}e_l^2+\frac{\lambda}{2}\left(\|\boldsymbol{\alpha}\|^2+\|\psi\|_{\mathcal{H}}^2\right)\\&-\sum_{l=1}^{L}\beta_l\left(e_l-x_l+\mathbf{d}_{\lambda_l}^T\boldsymbol{\alpha}+\psi\left(\mathbf{d}_{\lambda_l}\right)\right)\\&-\sum_{d=1}^{N}\mu_d\alpha_d+\nu\left(\mathbf{1}^T\boldsymbol{\alpha}-1\right)\end{aligned}\tag{19}$$

where β_l, μ_d, and ν are Lagrange multipliers. The optimization can be achieved by minimizing L, and the conditions for optimality with respect to the primal variables are

$$\begin{cases}\boldsymbol{\alpha}^*=\dfrac{1}{\lambda}\left(\displaystyle\sum_{l=1}^{L}\beta_l^*\mathbf{d}_{\lambda_l}+\boldsymbol{\mu}^*-\nu^*\mathbf{1}\right)\\\psi^*=\dfrac{1}{\lambda}\left(\displaystyle\sum_{l=1}^{L}\beta_l^*k\left(\cdot,\mathbf{d}_{\lambda_l}\right)\right)\\e_l^*=\beta_l^*.\end{cases}\tag{20}$$

By substituting (20) into (19), we can get the following dual problem

$$\begin{aligned}&\max_{\boldsymbol{\beta},\boldsymbol{\mu},\nu}L'(\boldsymbol{\beta},\boldsymbol{\mu},\nu)\\&=-\frac{1}{2\lambda^2}\begin{bmatrix}\boldsymbol{\beta}\\\boldsymbol{\mu}\\\nu\end{bmatrix}^T\begin{bmatrix}\mathbf{K}+\lambda\mathbf{I}&\mathbf{D}&-\mathbf{D}\mathbf{1}\\\mathbf{D}^T&\mathbf{I}&-\mathbf{1}\\-\mathbf{1}^T\mathbf{D}^T&-\mathbf{1}^T&N\end{bmatrix}\begin{bmatrix}\boldsymbol{\beta}\\\boldsymbol{\mu}\\\nu\end{bmatrix}\\&\quad+\frac{1}{\lambda}\begin{bmatrix}\mathbf{x}\\\mathbf{0}\\-1\end{bmatrix}^T\begin{bmatrix}\boldsymbol{\beta}\\\boldsymbol{\mu}\\\nu\end{bmatrix}\\&\mathbf{s.t.}\ \boldsymbol{\mu}\geq 0\quad\text{with}\quad\mathbf{K}=\mathbf{D}\mathbf{D}^T+\mathbf{K}_{\text{nlin}}.\end{aligned}\tag{21}$$

Problem (21) is a quadratic programming, which can be solved by interior point, activate set, or conjugate gradient algorithms [51]. Once the problem is solved, we can get the representation coefficient vector $\boldsymbol{\alpha}=\left[\boldsymbol{\alpha}_b^T,\ \boldsymbol{\alpha}_a^T\right]^T$ as follows:

$$\boldsymbol{\alpha}^*=\frac{1}{\lambda}\left(\mathbf{D}^T\boldsymbol{\beta}^*+\boldsymbol{\mu}^*-\nu^*\mathbf{1}\right).\tag{22}$$

Then, the anomalous degree of the test pixel can be determined by the elements in $\boldsymbol{\alpha}_a$ as

$$\|\boldsymbol{\alpha}_a\|_2=\sqrt{\sum_{i=1}^{R}\alpha_{a_i}^2}.\tag{23}$$

IV. Experimental Results and Analysis

In the experiments, we utilize one synthetic dataset and three real hyperspectral datasets to evaluate the proposed method. A comprehensive analysis about the proposed KNUD is first conducted on the synthetic dataset, which is generated with the definite ground-truth map. Then, the three real datasets are used to validate the effectiveness of the proposed KNUD in practical applications.

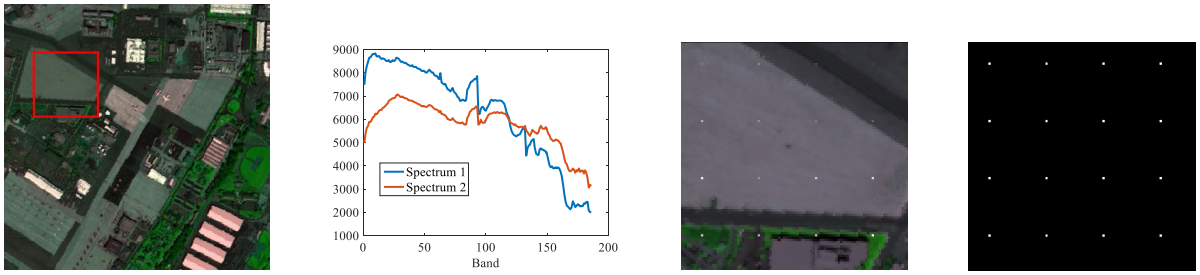
We compare the detection performance of the proposed KNUD with the conventional as well as the state-of-the-art anomaly detectors, including GRX [7], LRX [8], CRD [16], KCRD [16], and KRX [25]. All of the detectors can be utilized to detect the anomalous targets at full-pixel and subpixel levels. The GRX and LRX are two versions of the RX detector, which are known as the benchmark of the anomaly detection method. CRD is a typical linear representation-based detector, and KCRD is the kernel version of the CRD, which maps pixels into a high-dimensional feature space. KRX is a nonlinear detector by computing the Mahalanobis distance in the high-dimensional feature space. For the proposed method KNUD, we use the second-order polynomial kernel and Gaussian kernel to individually model the nonlinear interactions among the dictionary atoms and denote them as KNUD (P) and KNUD (G), respectively. The color detection maps of the different detectors are provided to compare the detection performance qualitatively. Besides, the receiver operating characteristic (ROC) curve and the area under the ROC curve (AUC) are utilized for the quantitative comparison. Moreover, the ROC curve with pointwise confidence bounds is utilized to compare the detection performance statistically as suggested in [52].

The experimental platform is a personal computer with an Intel Core i7-6700K CPU at 4.00 GHz and 32-GB RAM. All the experiments are implemented on MATLAB R2015b.

A. Experiments on Synthetic Dataset

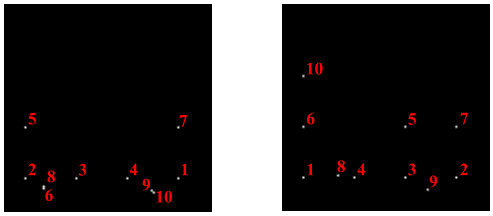
The synthetic dataset is generated based on the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) dataset, which is collected from San Diego airport, CA, USA, with a spatial resolution of 3.5 m. It contains 224 spectral bands in the wavelengths ranging from 370 to 2510 nm. In our experiments, 186 bands are preserved after removing the water absorption and low signal-to-noise ratio (SNR) bands (1–6, 33–35, 94–97, 107–113, 153–166, and 221–224). The original whole dataset has an image scene of 400 × 400 pixels, as shown in Fig. 3(a). We select a subimage with 100 × 100 pixels in the red square area to construct the synthetic dataset. Here, we refer to [49] and [53] for the target implantation. The embedded anomalous targets $\mathbf{t}$ are selected from two different airplanes in the original scene, the spectral signatures of which are shown

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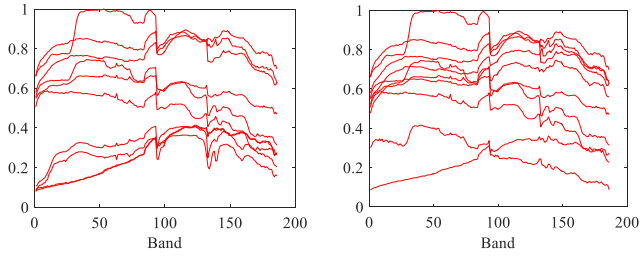
(a) (b) (c) (d)

Fig. 3. Synthetic hyperspectral dataset. (a) Pseudocolor image of the whole scene. (b) Spectral signatures of the selected anomalies **t**. (c) Pseudocolor image of the detection region. (d) Ground-truth map.



(a) (b)

Fig. 4. Selected pixels with order for the construction of the potential anomaly dictionary by different initial detectors. (a) GRX. (b) KRX.



(a) (b)

Fig. 5. Spectra of the selected pixels for the construction of the potential anomaly dictionary by different initial detectors. (a) GRX. (b) KRX.

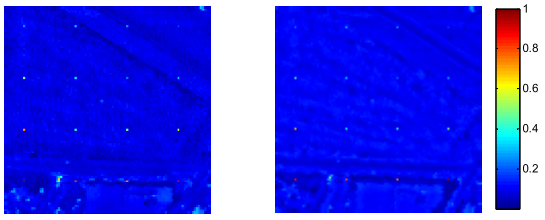
in Fig. 3(b). The synthetic anomalous pixel **z** is generated by fractionally implanting anomaly **t** into the originally given background pixel **b** at the given position according to PPNM as

$$\mathbf{z} = f \cdot \mathbf{t} + (1 - f) \cdot \mathbf{b} + \xi f(1 - f) \cdot (\mathbf{t} \odot \mathbf{b}) \tag{24}$$

where f is the mixing coefficient, which takes the value of 0.05, 0.1, 0.2, and 0.4 for each row from top to bottom, and ξ is a random value to denote the weight of the nonlinear part. Target spectrum 1 is embedded in the left two columns, and target spectrum 2 is embedded in the right two columns. The pseudocolor image and ground-truth map of the synthetic dataset are shown in Fig. 3(c) and (d), respectively.

1) Construction of the Potential Anomaly Dictionary: Here, we first analyze the influence of the initial detection map for the construction of the potential anomaly dictionary. As mentioned in Section III-A, we choose R pixels as the potential anomaly dictionary atoms based on the initial detection map. The parameter of the anomaly dictionary atom number is empirically set to $R = 10$. We use linear detector GRX and nonlinear detector KRX to obtain the initial detection maps respectively, and then compare the final detection performance of the proposed KNUD (P).

The selected pixels with order for constructing the potential anomaly dictionary by different initial detectors are shown in Fig. 4(a) and (b). Also, the spectra of these selected pixels are plotted in Fig. 5(a) and (b). It can be observed that the spectra of the potential anomalous atoms selected by different initial detectors are similar. Besides, it is noteworthy that there are some background pixels belonging to sparse materials mistakenly selected to be the anomalous pixels according to Fig. 3(d). Nevertheless, the aim of the anomaly dictionary is to find the potential anomaly, and the global false alarms can be suppressed by the utilization of the background dictionary, which pays more attention to the local region.



(a) (b)

Fig. 6. 2-D color detection results obtained by the proposed KNUD (P) with potential anomaly dictionaries constructed by different initial detectors. (a) GRX. (b) KRX.

Furthermore, the detection results obtained by the proposed KNUD (P) with different anomaly dictionaries are shown in Fig. 6. The background dictionary is constructed by the dual window with the size $(W_{\text{in}}, W_{\text{out}}) = (3, 11)$, and η is set to 95. The tradeoff parameter is set to $\lambda = 1$. The parameter settings will be analyzed in Section IV-A2. It can be observed from Fig. 6 that the color detection maps are similar although they use different detectors to construct the anomaly dictionary in the proposed method. The false alarms that occur in the initial detection map are effectively suppressed. Moreover, the AUC value is utilized to compare the detection performance quantitatively, and they are 0.9989 and 0.9992 for the anomaly dictionary constructed from GRX and KRX, respectively. However, KRX takes 315.8336 s to obtain the initial detection map, whereas GRX only takes 0.8020 s. As a result, for efficiency, we use GRX to generate the initial detection map and further for the construction of the anomaly dictionary.

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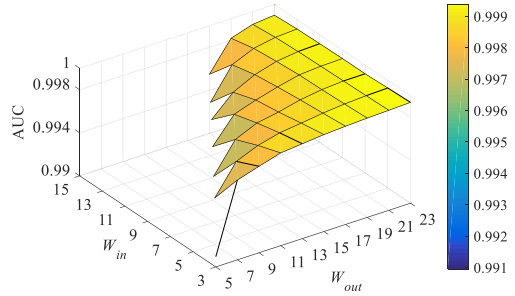


Fig. 7. Influence of the parameters (W_{in}, W_{out}) on the detection performance of the proposed method.

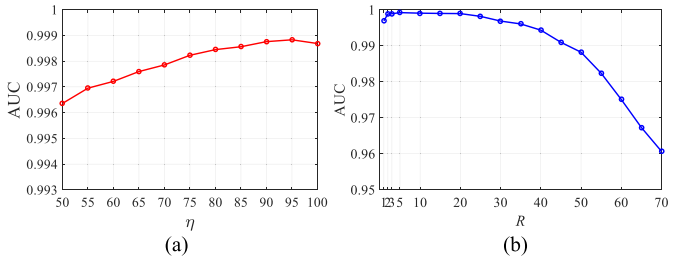


Fig. 8. Influence of the parameters in the construction of union dictionary on the detection performance of the proposed method. (a) Percent of the pixels selected in the dual window ($\eta\%$) to construct the background dictionary and (b) number of the atoms (R) in the potential anomaly dictionary.

2) Parameter Settings: In this section, we analyze the influence of the parameters involved in the proposed method on the detection performance. Here, we employ the second-order polynomial kernel for the proposed KNUD in the experiments.

First, we investigate the influence of the parameters in the union dictionary construction, i.e., the size of the dual window (W_{in}, W_{out}), the η percent of the pixels selected in the dual window to construct the background dictionary, and the number of anomaly dictionary atoms R. Figs. 7 and 8 show the impacts of these parameters on the proposed method in terms of the AUC values. When analyzing one parameter, the others are fixed. As shown in Fig. 7, the AUC value decreases with the increase of W_{in} when W_{out} is fixed, and the AUC value increases with the increase of W_{out} when W_{in} is fixed. This can be explained by that each anomalous pixel is distributed dispersedly in the synthetic dataset. It can also be observed that the detection performance is stable in a wide range of parameters W_{in} and W_{out}. Generally, (W_{in}, W_{out}) are set according to the real hyperspectral dataset, and W_{in} is larger than the size of the potential anomalous objects [46], [54]. This is due to that a smaller W_{in} may result in serious contamination of the background dictionary and thereby may decrease the detection accuracy. From Fig. 8(a), we can see that the detection performance in terms of AUC values first gradually increases and the optimal result is achieved when $\eta = 95$. However, if we set $\eta = 100$, which means selecting all pixels in the dual window to construct the background dictionary, the detection accuracy decreases. This indicates that the purification of the background dictionary is meaningful to the proposed KNUD, and therefore, we set $\eta = 95$ throughout the experiments. Fig. 8(b) shows the variation of detection accuracy with regard to AUC values under different choices of R. The proposed KNUD can obtain the satisfactory

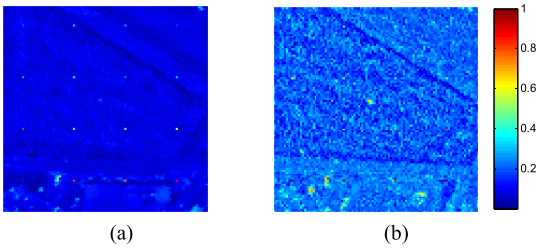


Fig. 9. 2-D color detection results obtained by the proposed KNUD (P) (a) with regularization term ($\lambda = 1$) and (b) without regularization term.

TABLE I
AUC VALUES OF THE DETECTION RESULTS OBTAINED BY KNUD (P) WITH DIFFERENT VALUES OF THE TRADEOFF PARAMETER λ

λ	0	0.001	0.01	0.1	1	2	3	5
AUC	0.8255	0.9675	0.9876	0.9986	**0.9989**	0.9988	0.9987	0.9984

detection results with few potential anomaly atoms, even when we choose only two or three of the strongest anomalous pixels (which can cover the two kinds of anomalous spectra) to construct the potential anomaly dictionary. This is because the purpose of constructing the potential anomaly dictionary is to find the strong anomalous pixels, and these pixels serve as the prior to represent other anomalies in the scene. Besides, the spectral signatures of the strong anomalous pixels are very different from the background, and these pixels can be easily identified by general anomaly detectors. When R increases, the potential anomaly dictionary will contain more sparse background pixels, which results in the decrease of the detection accuracy. Overall, the proposed KNUD can achieve satisfactory detection results with a relatively wide range of R. Considering the fact that the real hyperspectral data may contain several kinds of anomalous materials and the number of the atoms in the potential anomaly dictionary should be larger than the kinds of the anomalous materials, we fix $R = 15$ in the real situations for simplicity.

Next, we analyze the influence of the tradeoff parameter λ on the final detection result. The tuning parameter is selected in the range [0, 0.001, 0.01, 0.1, 1, 2, 3, 5], and the detection performance evaluated by AUC is reported in Table I. It can be seen that the value of λ can be selected from a relatively wide range and an appropriate setting can be $\lambda = 1$. Moreover, Fig. 9(a) and (b) shows the detection results of the proposed KNUD (P) with and without regularization term, respectively. It can be observed that the regularization term is useful at keeping the spatial smoothness of the background and highlight anomalous pixels in the detection map. This can be attributed to the effectiveness of the regularization term in avoiding overfitting.

3) Comparison With Other Anomaly Detectors: We compare the detection performance of the proposed method KNUD (P) and KNUD (G) with GRX, LRX, CRD, KCRD, and KRX using the synthetic dataset. The parameters involved in the compared detectors are optimally tuned or set according to the recommended values in the literature. Specifically, the size of the dual window (W_{in}, W_{out}) is set to (3, 11) for LRX, CRD, KCRD, and the proposed KNUD. The tradeoff parameter λ in CRD and KCRD

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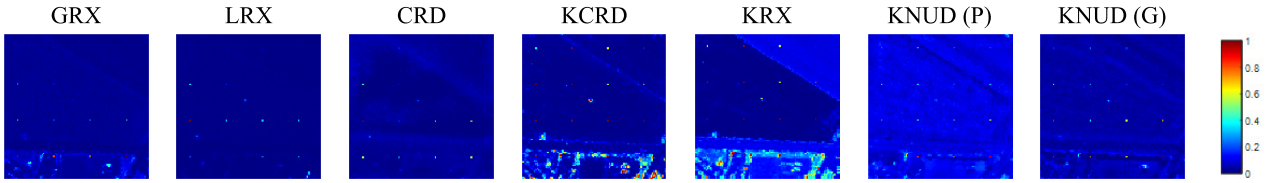


Fig. 10. 2-D color detection results of different detectors on the synthetic dataset.

is set to 10^{-6} as a default setting suggested in [16]. For KRX, we select the Gaussian kernel for the nonlinear mapping, and the bandwidth of the kernel function is $\sigma = 1$, which is the same as KCRD and the proposed KNUD (G).

The 2-D color detection results of different detectors are presented in Fig. 10. It can be observed that all detectors can identify the anomalous pixels with relatively high mixing coefficients $f = 0.4$ and $f = 0.2$. However, when the mixing coefficient is lower just as the weak anomalous pixels in the first row, the GRX, LRX, and CRD are unable to well discriminate the anomalies. This indicates that the discrimination power of the linear anomaly detectors is not effective when the anomalous object is mixed with the background in a lower abundance. In addition, KCRD and KRX perform well in separating the weak anomalies from the background. However, in the bottom of the detection maps, there are a considerable number of false alarms of the background. This is because KCRD and KRX simply replace the inner product of the pixels that applied in the linear algorithms by the kernel function. The kernel function can implicitly map the pixel into a high-dimensional feature space and increase the separability of the pixels. However, this strategy does not take the intrinsic nonlinear characteristics into account, and the inappropriate nonlinear mapping will make background pixels more dispersed, which results in high false alarms. Overall, the proposed KNUD (P) and KNUD (G) provide the best detection results, and they can effectively identify all of the anomalous pixels and suppress the false alarm in the bottom region. This is because the proposed KNUD takes the hyperspectral nonlinear mixing models into consideration and deals with the intrinsic nonlinear characteristics by utilizing the nonlinear interaction among the dictionary atoms to design the representation-based detector. Besides, dual window is effective to exploit the spatial homogeneity and suppress the false alarms of the background, which can also be verified by LRX and CRD detectors.

To evaluate the detection performance quantitatively, the ROC curves of different detectors are shown in Fig. 11. It shows that the proposed KNUD (P) and KNUD (G) can achieve a 100% detection rate with lower false alarm rate than the other detectors. Besides, the confidence bounds of the proposed method are higher than the compared detectors at almost all false alarm rates, which shows the statistically significant improvement of the proposed method over the competitors. Moreover, the AUC values are reported in Table II. The proposed KNUD obtains higher AUC values than the others, which indicates the superiority of the proposed method. Specifically, the detection performance of KNUD (P) is better than that of KNUD (G). It is because we generate the synthetic dataset based on the PPNM, which is widely used to model the nonlinear mixing by considering the second-order scattering effect [36].

TABLE II
AUC VALUES AND RUNNING TIME OF DIFFERENT ANOMALY DETECTORS ON THE SYNTHETIC DATASET

Algorithm	GRX	LRX	CRD	KCRD	KRX	KNUD (P)	KNUD (G)
AUC	0.9556	0.9854	0.9870	0.9935	0.9899	**0.9989**	0.9979
Time (s)	0.7619	0.8931	12.8212	156.8122	320.7105	203.7017	205.0851

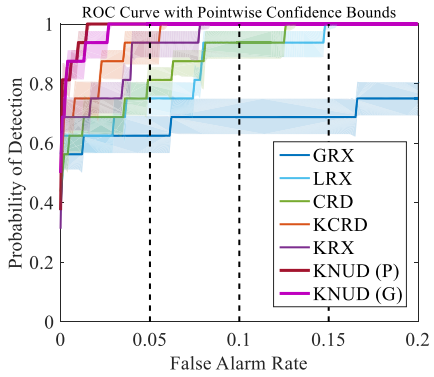


Fig. 11. ROC curves of different detectors on the synthetic dataset.

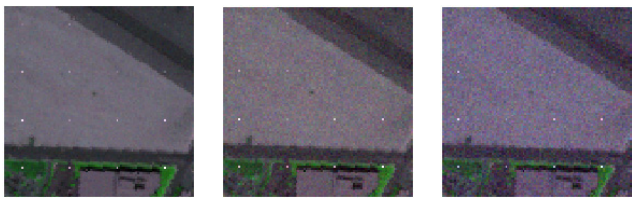


Fig. 12. Pseudocolor images of the synthetic dataset with different levels of zero-mean white Gaussian noise. (a) 40 dB. (b) 35 dB. (c) 30 dB.

As for the running time of different detectors reported in Table II, the linear methods cost less time to obtain the detection results in comparison with the nonlinear methods. This is because the computational cost of forming the kernel matrix is heavy and the operations on kernel matrix, such as singular value decomposition (SVD), are also time-consuming. The running time of the KRX is longer than those of KCRD and the proposed method. This is because KRX computes the kernel matrix of the whole HSIs, whereas the KCRD and the proposed method compute the kernel matrix in the local region. Besides, a closed-form solution of KCRD is provided in [16], but the proposed method needs to solve an extra optimization problem. As a result, our proposed method costs a little longer time than KCRD.

In summary, the experimental results demonstrate that the proposed KNUD outperforms the other linear and nonlinear

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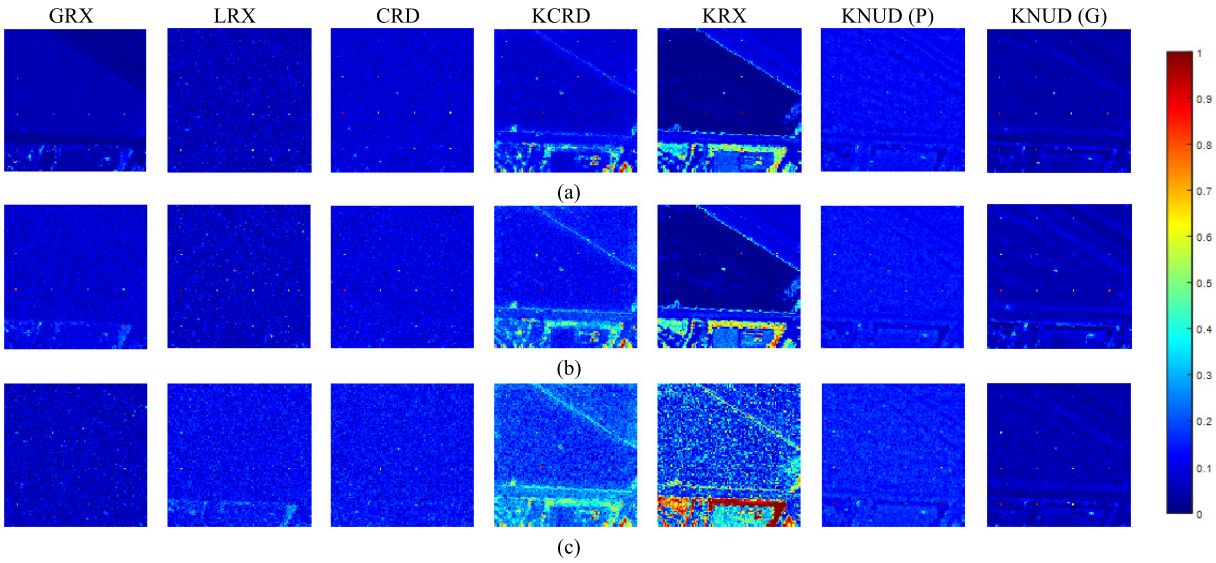


Fig. 13. 2-D color detection results of different detectors on the synthetic dataset with different levels of zero-mean white Gaussian noise. (a) 40 dB. (b) 35 dB. (c) 30 dB.

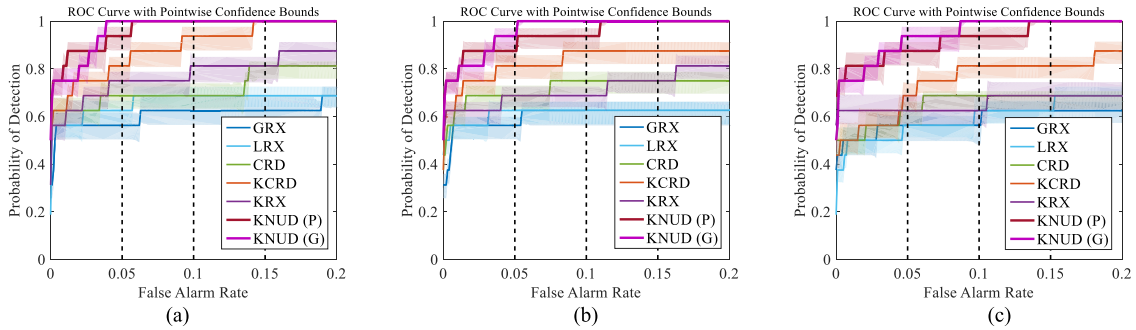


Fig. 14. ROC curves of different detectors on the synthetic dataset with different levels of zero-mean white Gaussian noise. (a) 40 dB. (b) 35 dB. (c) 30 dB.

anomaly detectors and the time cost is also in an acceptable range. Moreover, the parallel computation and the GPU are possible ways to cut down the time cost of the kernel-based methods in practical applications.

4) Robustness on the Situations With Noise: In order to further evaluate the capability and stability of these detection methods on the situations with noise, we corrupt the synthetic dataset with different levels of zero-mean Gaussian noise. The SNR is set to 40, 35, and 30 dB, Here, the SNR is defined as follows:

$$\mathrm{SNR} \equiv 10\log_{10}\frac{\mathbb{E}\left[\mathbf{y}^T\mathbf{y}\right]}{\mathbb{E}\left[\mathbf{e}^T\mathbf{e}\right]} \tag{25}$$

where $\mathbb{E}[\cdot]$ stands for the expectation operator, $\mathbf{y}$ and $\mathbf{e}$ denote the spectral signature and the additive noise of a pixel, respectively.

The pseudocolor images of the synthetic dataset with different levels of noise are shown in Fig. 12. The 2-D color detection maps of different detectors are presented in Fig. 13. It can be observed that the detection performance of the proposed method is much more effective and stable than the compared detectors. The corresponding ROC curves of different detectors on the synthetic dataset under different levels of noise are shown in Fig. 14. The proposed KNUD (P) and KNUD (G) have higher detection probability and lower false alarm rate.

Then, we perform these detectors 20 times with randomly distributed noise to investigate the robustness. The average AUC values with the standard deviations (std) are reported in Table III. It can be seen that KNUD (P) and KNUD (G) show the high accuracy and robustness on the situations with noise. Specially, KNUD (G) obtains the highest AUC values on the all situations, which can be explained by that the added Gaussian noise may influence the original nonlinear characteristics and, thus, the second-order polynomial kernel may be insufficient to model the nonlinear interactions among the atoms in the union dictionary. On the contrary, the Gaussian kernel has the ability to extract the higher-order nonlinear interactions, and it could be concluded that KNUD (G) is more robust in noisy situations.

B. Experiments on Real Hyperspectral Datasets

This section evaluates the practical effectiveness of the proposed KNUD by utilizing three widely used real hyperspectral datasets, which are collected from different sensors. The background and anomaly of the real datasets are distributed nonuniformly, and the regions selected in the experiments are the urban and airport area, which are generally influenced by

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TABLE III

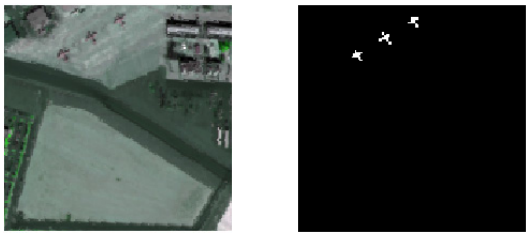
AUC VALUES (MEAN ± STD) OF DIFFERENT DETECTORS ON THE SYNTHETIC DATASET WITH DIFFERENT LEVELS OF ZERO-MEAN WHITE GAUSSIAN NOISE

Algorithm	GRX	LRX	CRD	KCRD	KRX	KNUD (P)	KNUD (G)
40 dB	0.9267±0.0228	0.9565±0.0161	0.9604±0.0146	0.9851±0.0083	0.9751±0.0119	0.9939±0.0018	**0.9967±0.0014**
35 dB	0.8911±0.0370	0.9156±0.0314	0.9453±0.0205	0.9785±0.0108	0.9669±0.0167	0.9916±0.0038	**0.9953±0.0021**
30 dB	0.8666±0.0421	0.8778±0.0415	0.9116±0.0214	0.9629±0.0211	0.9150±0.0235	0.9865±0.0068	**0.9924±0.0030**



(a) (b) (c)

Fig. 15. HYDICE dataset. (a) Pseudocolor image of the whole dataset. (b) Pseudocolor subimage of the detection area. (c) Ground-truth map of the anomalous pixels.



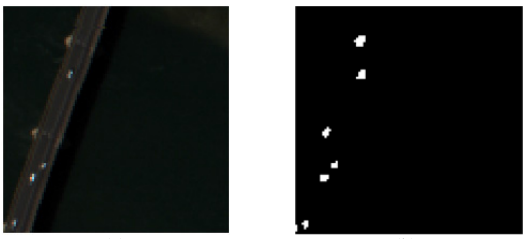
(a) (b)

Fig. 16. AVIRIS dataset. (a) Pseudocolor image of the detection area. (b) Ground-truth map of the anomalous pixels.

the multiple light scattering effects, and some materials are nonlinearly mixed in a pixel.

The first dataset was collected by the HYperspectral Digital Imagery Collection Experiment (HYDICE) airborne sensor as shown in Fig. 15(a), which covers an urban residential area. The original HSIs contain 307 × 307 pixels and 210 spectral bands in the wavelength range of 400–2500 nm, and its spatial resolution is 1 m. In the experiments, 160 bands are retained after eliminating the water vapor absorption and low-SNR bands (1–4, 76, 87, 101–112, 136–153, and 197–210). Then, a subimage with 80 × 100 pixels in the upper right area of the original image is used in the experiments. The anomalous objects to be detected are the vehicles and roofs, and some anomalies appear in sub-pixel level due to their small sizes [55]. The pseudocolor image and the ground-truth map are presented in Fig. 15(b) and (c), respectively.

The second real HSIs used in the experiment are also a portion of AVIRIS. The upper left area with 150 × 150 pixels is used for anomaly detection in the practical situation, as shown in Fig. 16(a). This scene has different types of background materials, including roads, roofs, grasses, and shadows. The anomalous objects to be detected are three airplanes, and the ground-truth map is shown in Fig. 16(b). In the edge of the airplanes, the spectra of the anomaly and backgound are mixed in one pixel. The sizes of the anomalies in this dataset are relatively large.



(a) (b)

Fig. 17. ROSIS dataset. (a) Pseudocolor image of the detection area. (b) Ground-truth map of the anomalous pixels.

TABLE IV

AUC VALUES AND RUNNING TIME OF DIFFERENT ANOMALY DETECTORS ON THE THREE REAL DATASETS

Algorithm		GRX	LRX	CRD	KCRD	KRX	KNUD (P)	KNUD (G)
HYDICE	AUC	0.9860	0.9879	0.9877	0.9918	0.9930	**0.9979**	0.9964
	Time (s)	0.5550	8.3723	10.7962	116.9754	193.8974	178.5349	179.1532
AVIRIS	AUC	0.9565	0.9382	0.9707	0.9883	0.9792	**0.9986**	0.9967
	Time (s)	2.6253	20.3298	57.1508	659.5752	1047.8326	622.1191	631.2355
ROSIS	AUC	0.9834	0.9828	0.9895	0.9957	0.9933	0.9978	**0.9984**
	Time (s)	0.7050	6.2755	9.2331	14.6754	319.4587	38.9683	40.1856

The third one was acquired by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor. This dataset was obtained during a flight campaign over Pavia, Northern Italy, with 102 bands and its geometrical resolution is 1.3 m. The original whole image contains 1096 × 1096 pixels, and a 100 × 100 subscene with an available ground-truth map is generally utilized for anomaly detection. In this dataset, the vehicles on the bridge are regarded as anomalies. Fig. 17(a) and (b) shows the pseudocolor image and the ground-truth map of the ROSIS dataset, respectively. We can see that the pixels appearing in the edge of the objects are sub-pixel anomalies.

For the HYDICE dataset, the size of dual window is set to $(W_{\text{in}},\ W_{\text{out}}) = (3,\ 11)$ in the proposed method, and these parameters in LRX, CRD, and KCRD are set to (3, 7), (3, 9), and (3, 9), respectively, after tuning. The 2-D color detection results obtained by different detectors are shown in Fig. 18(a). It can be observed that some anomalous pixels cannot be clearly identified in the detection maps of linear detectors. KCRD and KRX can locate almost all anomalies. However, we can see that some background pixels also exhibit high response values in the detection results, which results to the high false alarm. The proposed KNUD (P) and KNUD (G) can not only effectively detect the anomalous pixels but also suppress the background. Specially, the detection result of KNUD (P) is better than that of KNUD (G), and this is mainly due to that the second-order scattering generally occurs in this area and the second-order polynomial kernel

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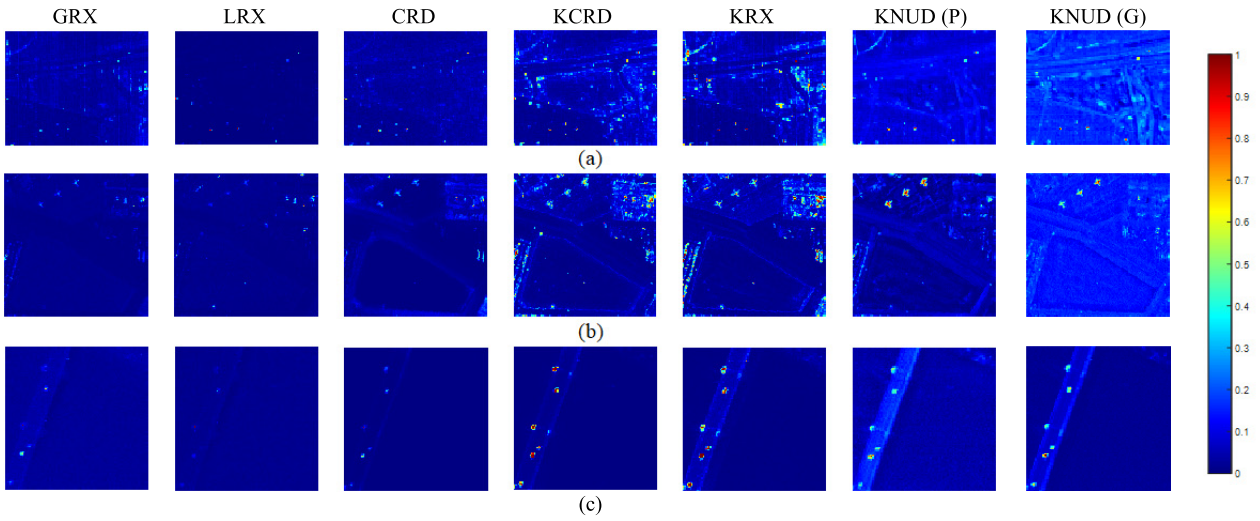


Fig. 18. 2-D color detection results of different detectors on the three real hyperspectral datasets. (a) HYDICE. (b) AVIRIS. (c) ROSIS.

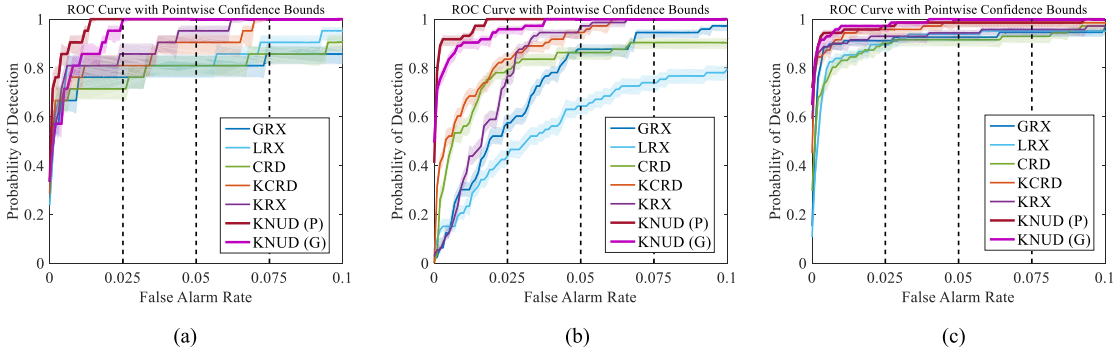


Fig. 19. ROC curves of different detectors on the three real hyperspectral datasets. (a) HYDICE. (b) AVIRIS. (c) ROSIS.

can better model the nonlinear interactions on this situation. To quantitatively compare the performance of these detectors, the ROC curves and AUC values are presented in Fig. 19(a) and Table IV, respectively. It can be seen from the ROC curves that KNUD (P) obtains the highest probability of detection on almost all different false alarm rates. Besides, KNUD (G) exhibits the lowest false alarm rate when the detection probability reaches 100%. The AUC values also show that the proposed methods outperform the other detectors. These results confirm the superiority of exploring the nonlinear interactions of the pixels in the proposed method.

For the AVIRIS dataset, anomaly detection is relatively more difficult due to the distribution of roofs and the sizes of the anomalous targets are large. In this dataset, we set the optimal dual window sizes (W_{in}, W_{out}) for LRX, CRD, KCRD, and the proposed methods to (9, 17), (11, 19), (9, 17), and (9, 15), respectively. Fig. 18(b) shows the color detection results of different detectors. It is obvious that the proposed methods perform much better than the other detectors in highlighting the airplanes and suppressing the background. Especially in the bottom-left and top-right areas, the false alarms caused by background pixels are effectively suppressed in the results of KNUD (P) and KNUD (G) compared with those of KCRD and KRX. Similar conclusions can be drawn from the ROC curves and the AUC values, as presented in Fig. 19(b) and Table IV, respectively.

For the ROSIS dataset, the dual window sizes (W_{in}, W_{out}) in LRX, CRD, KCRD, and the proposed methods are set to (5, 9), (7, 13), (7, 11), and (7, 13), respectively, after extensive searching. The color detection maps are shown in Fig. 18(c). Clearly, the anomalous objects in the results of nonlinear detectors are visually more distinct from the background, whereas the linear detectors are not effective to identify the anomalies. The ROC curves shown in Fig. 19(c) also validate the better performance of the nonlinear detection methods. Moreover, the KNUD (G) obtains the highest AUC value, and this can be explained by the fact that the vehicles, the bridge, and the river are in different levels; the high-order scattering effects result in the multilayer nonlinear mixing of the materials on the ground surface.

Furthermore, we compare the computation time of the different detectors on the three real datasets, which is also summarized in Table IV. It can be seen that the linear detectors consume less time compared with the nonlinear detectors, and the speed of GRX is the fastest because it simply computes the Mahalanobis distance to detect the anomalies. By comparing the speed of CRD and KCRD, GRX, and KRX, we can conclude that kernel methods cost more time. Moreover, the proposed KNUD (P) and KNUD (G) are faster than KRX since

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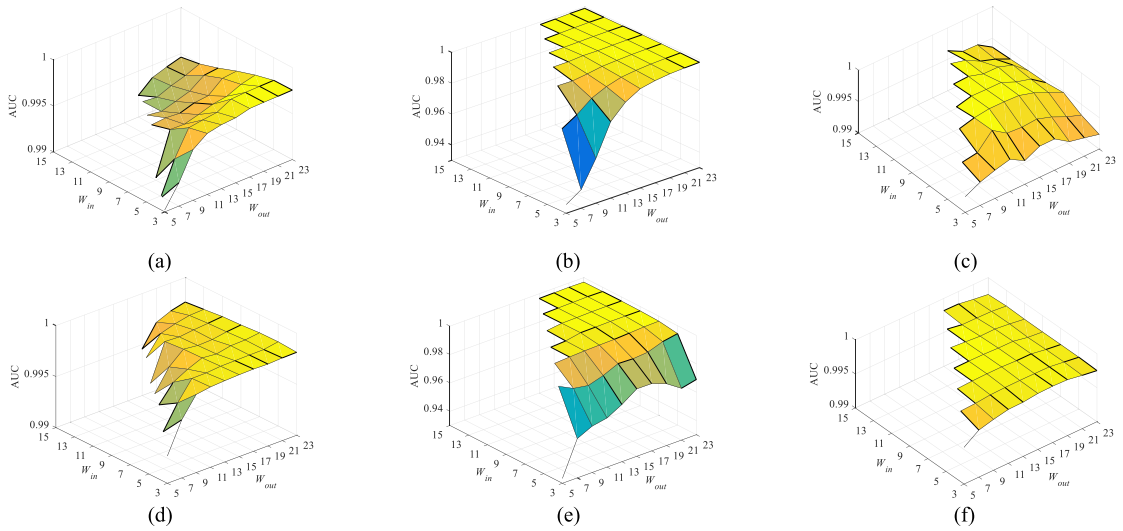


Fig. 20. Influence of the parameters (W_{in}, W_{out}) on the detection performance of the KNUD (P) for the three real hyperspectral datasets: (a) HYDICE, (b) AVIRIS, and (c) ROSIS, and KNUD (G) for the three real hyperspectral datasets: (d) HYDICE, (e) AVIRIS, and (f) ROSIS.

they utilize the dual window and compute the kernel matrix in the local region rather than the global HSIs.

Finally, we analyze the influence of the parameters (W_{in}, W_{out}) on the proposed method for the three real hyperspectral datasets. We set the sizes of the dual window according to the dataset for the proposed method. This is because the sizes of the objects in the HSIs are influenced by spatial resolution. Besides, the irregular distribution of the objects may result in serious contamination of the background dictionary. As shown in Fig. 20, we can see that the proposed method can obtain satisfactory detection results in a wide range of parameters (W_{in}, W_{out}). This can be attributed to the employed strategy of refining the background dictionary by calculating the Mahalanobis distance.

C. Summary

According to the experimental results on the synthetic dataset and the three real hyperspectral datasets, we can summarize the advantages of the proposed method as follows.

1) *Effectiveness:* The proposed KNUD utilizes the nonlinear combination of the atoms in the union dictionary to represent each test pixel. Based on the nonlinear mixing models (PPNM and MLM), the nonlinear interactions among the pixels are implicitly achieved by the kernel theory. This strategy is more appropriate and physically interpretable to model the intrinsic nonlinear characteristics in the HSIs. The experimental results demonstrate that the proposed KNUD is effective to distinguish the anomalous target from the background and maintains a very low false alarm rate when the detection probability reaches 100%.
2) *Simplicity of Parameter Settings:* The analysis on the selection of parameters indicates that the detection performance of our proposed method is robust to η and R, which are utilized to construct the union dictionary and fixed throughout the experiments. Besides, the tradeoff parameter is set to $\lambda = 1$ since the detection performance is insensitive to this parameter. We change the dual window sizes (W_{in}, W_{out}) according to the size of the potential anomalous objects, and the experimental results show that the satisfactory detection results can be obtained in a wide selection range of (W_{in}, W_{out}).
3) *Robustness to the Noise:* Different levels of Gaussian noise are added to the synthetic dataset to evaluate the detection performance on the noisy situation, and the experimental results show that the proposed KNUD can obtain the satisfactory detection results even when the SNR is very low.

V. Conclusion

In this article, we have presented a novel nonlinear anomaly detection method named KNUD. In comparison with other nonlinear anomaly detectors, the proposed method takes the nonlinear mixing models into consideration to model the intrinsic nonlinear characteristics of the real HSIs, which is more physically interpretable. A union dictionary consisting of background and anomalous atoms is constructed by considering the local spatial correlations and global anomalies. With the union dictionary and the representation-based theory, each test pixel is represented by the nonlinear combination of the atoms in the union dictionary. Instead of simply mapping each pixel into a high-dimensional feature space, the proposed KNUD considers the nonlinear interactions among the atoms in the union dictionary, which relies on the nonlinear mixing models with a clearly physical interpretation. Finally, the anomalies are accurately determined by means of the representation coefficients of the anomaly dictionary. The experimental results on both synthetic and real hyperspectral datasets have verified the effectiveness, robustness, and stability of the proposed method, and the proposed KNUD can provide excellent detection performance in comparison with the other linear and nonlinear anomaly detectors.

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In the future research, we would study the multiple kernel learning strategy to better model the intrinsic nonlinear characteristics of the specific HSIs scene for anomaly detection.

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